Fiscal Year:	FY 2016	**Task Last Updated:**	FY 06/27/2018
PI Name:	Ferri, James Ph.D.		
Project Title:	Particle Stabilized Emulsions and Foams		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	23219	**Congressional District:**	4
Comments:	NOTE: Previous affiliation was Lafayette College, Easton, PA until early 2018 when PI moved to Virginia Commonwealth University		
Project Type:	Ground	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	03/26/2015	**End Date:**	08/15/2017
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX15AE98G ; NNX10AV26G		
Performance Goal No.:			
Performance Goal Text:			
	ED. NOTE: Continues research from grant NNX10AV26G, with same title. Overall project goal: Design and synthesize stimulus responsive nanoparticles using thermoresponsive polymer grafts to provide switchable control over surface chemistry and macroscopic dispersion (foams and emulsions) stability. The goals of the research are to demonstrate a stimulus-responsive design of dispersion stability and a mechanistic decoupling of interfacial and colloidal effects associated with foam de/stabilization using conditions of microgravity. The project goals are divided into four tasks: Task 1: Stimulus-responsive (nano)particle design. Task description: Design and synthesize core-shell nanostructures		

	where 15 nm diameter gold nanoparticles (NPs) are covered by a shell of disulfide functionalized thermoresponsive copolymers of di(ethylene glycol) methyl ether methacrylate (MeO2MA) and poly(ethylene glycol) methyl ether methacrylate (OEGMA) via chemical surface reaction.
Task Description:	Task 2: Characterization of colloidal stability of stimulus-responsive (nano)particles. Task description: Nanoparticle colloidal stability is measured using time-resolveddynamic light scattering (TR-DLS). This method offers a means for direct measurement of k11, the initial rate constant for NP aggregation. Task 3: Characterization of adsorption and interfacial rheology of stimulus-responsive (nano)particles at air-water interface. Task description: Investigate the adsorption dynamics and interfacial rheology of Au@(MeO2MAx-co-OEGMAy) at the air-water interface as a function of external field (temperature). Task 4: Assessing foam/emulsion stability/destabilization in response to external field and environmental conditions. Task description: We seek to characterize modulation of interfacial properties, e.g., interfacial tension, surface elasticity, and surface viscosity, of oil-in-water emulsions by external fields and environmental conditions using thermoresponsive particulate systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	[Ed. note: compiled from 2016 annual report received May 2018] The project goals are divided into four tasks as follows and Progress Status is listed as well: Task 1: Stimulus-responsive (nano)particle design. Task description: Design and synthesize core-shell nanostructures where 15 nm diameter gold nanoparticles (NPs) are covered by a shell of disulfide functionalized thermoresponsive copolymers of di(ethylene glycol) methyl ether methacrylate (MeO2MA) and poly(ethylene glycol) methyl ether methacrylate (OEGMA) via chemical surface reaction. The disulfide copolymers, or DS-Poly(MEO2MAx-co-OEGMAy), are grown from a disulfide functionalized DTBE ATRP initiator, where x and y represented the molar fractions of MEO2MA and OEGMA, respectively. The water soluble DS-Poly(MEO2MAx-co-OEGMAy) display sharp property changes in response to temperature or ionic strength switches. In particular, they exhibit a lower critical solution temperature (LCST) in aqueous medium that can be tuned by changing the copolymer ratio. Status: Complete; Au@(MeO2MA90-co-OEGMA10), (MeO2MA80-co-OEGMA20) complete, (MeO2MA95-co-OEGMA5) have been synthesized and purified. Task 2: Characterization of colloidal stability of stimulus-responsive (nano)particles. Task description: Nanoparticle colloidal stability is measured using time-resolveddynamic light scattering (TR-DLS). This method offers a means for direct measurement of k11, the initial rate constant for NP aggregation. The increase in concentration of dimers, (C2) is proportional to the product of the initial rate constant of aggregation k11 and the square of the initial concentration of nanoparticle monomer (C1). Effects of temperature and ionic strength on aggregation kinetics is studied by dynamic light scattering (DLS). The kinetic and structural parameters were determined from the time evolution of the mean size and the scattering intensity, respectively. The hydrodynamic radius increase with time was monitored for a family of temperatures and NaCl concentrations; the initial growth rates were converted to the aggregation rate constant, k11. At higher temperatures lower salt concentrations induce aggregation – in a programmable manner. Extracting the NaCl concentration responsible for aggregation, i.e., the CAC, for each temperature, we can relate CAC, T, and x:y copolymer ratio. This suggests the colloidal aggregation and therefore stability can be programmed to respond to both external fields and environmental conditions. Additionally we have developed extended DLVO theory using fundamental physical chemistry to describe hydration potential inter-particle interaction potential and predict aggregation state. Deliverables: a.) experimental results relating nanoparticle concentration and solution chemistry to colloidal behaviour such as aggregation; b.) determination of the relationship between bulk interaction potential and surface chemistry. Status: Complete; a.) aggregation kinetics for (MeO2MA90-co-OEGMA10), (MeO2MA80-co-OEGMA20) complete, (MeO2MA95-co-OEGMA5) complete, b.) complete. Task 3: Characterization of adsorption and interfacial rheology of stimulus-responsive (nano)particles at air-water interface. Task description: Investigate the adsorption dynamics and interfacial rheology of Au@(MeO2MAx-co-OEGMAy) at the air-water interface as a function of external field (temperature). The surface tension change due to the adsorption of NPs was evaluated with respect to kinetic-diffusive transport models, to determine both the diffusion time and interparticle separation at the surface. The equilibrium and kinetics of adsorption indicate a strong sensitivity to both salt concentration (CNaCl<0.1M) and temperature (37<T<40ºC) suggesting a range of stimulus-responsive control of the interface is accessible. Deliverables: experimental results relating nanoparticle concentration and solution chemistry to a.) adsorption and b.) interfacial rheology. Status: Partially complete; a.) adsorption and interfacial rheology for (MeO2MA90-co-OEGMA10complete, a.) and b.) (MeO2MA80-co-OEGMA20) (MeO2MA95-co-OEGMA5) not yet complete. Task 4: Assessing foam/emulsion stability/destabilization in response to external field and environmental conditions. Task description: We seek to characterize modulation of interfacial properties, e.g., interfacial tension, surface elasticity, and surface viscosity, of oil-in-water emulsions by external fields and environmental conditions using thermoresponsive particulate systems. One of the primary challenges in the interfacial characterization of emulsions stabilized by thermoresponsive nanoparticles is the destabilization of the nanoparticle suspension upon the increase in hydrophobicity activated by a change in temperature or solution condition. Aggregated systems are susceptible to sedimentation, because of the density contrast between the suspending fluid and the aggregate. Thus, there is a dynamic competition between macroscopic emulsion destabilization, microscopic nanoparticle aggregation, and the interfacial transport processes that govern nanoparticle partitioning at the oil-water interface. A key aspect of interfacial transport and particle partitioning is the contact angle at which the particle attaches to the oil-water interface. We studied the contact angle as a function of temperature for a series of copolymers of varying compostion, and therefore temperature sensitivity. Microgravity remains an the ideal environment for decoupling nanoparticle colloidal dynamics from the processes associated with interfacial transport and emulsion destabilization. The FASTER facility can also be used for measuring interfacial tension, surface elasticity, and surface viscosity. Recently, FOAM-C has been proposed for use as a

	test frame for measuring macroemulsion dynamics and the transitions associated with emulsion destabilization. Deliverables: Study of foam destabilization in ground-based and microgravity conditions using FASTER and LIFT (for single bubble experiments) and FOAM-C for macroscopic foams. During 2015, the LIFT project was abandoned by the European Space Agency (ESA) and is now defunct. Project meetings for incorporating the activities of LIFT and PASTA (Particle Stabilized Emulsions and Foams) into FOAM-C began in November 2015 and will continue in 2016.
Bibliography Type:	Description: (Last Updated: 12/03/2024)
Articles in Peer-reviewed Journals	Gambinossi F, Defnet E, Ferri JK, Mylon SE. "Aggregation kinetics of stimulus-responsive polymer-coated gold nanoparticles driven by Hofmeister effects." Colloids and Interface Science Communications. 2015 Nov;9:9-11. https://doi.org/10.1016/j.colcom.2016.02.002 , Nov-2015
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Articles in Peer-reviewed Journals	Gambinossi F, Sefcik LS, Wischerhoff E, Laschewsky A, Ferri JK. "Engineering adhesion to thermoresponsive substrates: effect of polymer composition on liquid-liquid-solid wetting." ACS Appl Mater Interfaces. 2015 Feb 4;7(4):2518-28. https://doi.org/10.1021/am507418m ; PubMed PMID: 25569588 , Feb-2015
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Articles in Peer-reviewed Journals	Gambinossi F, Ferri JK. "Effect of electrolyte-induced polymer swelling on macromolecule functionalized core–shell nanoparticle interaction potential: Calculation of colloidal stability ratio." Colloids and Surfaces A: Physicochemical and Engineering Aspects. 2015 Jan 5;464:143-53. https://doi.org/10.1016/j.colsurfa.2014.10.018 , Jan-2015